

Supplementary diagnostics and implementation figures

Crowd Supervision and Vocabulary-Mismatched Selective Facial-Expression Prediction

SN Computer Science

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Interpretation and evidence boundaries

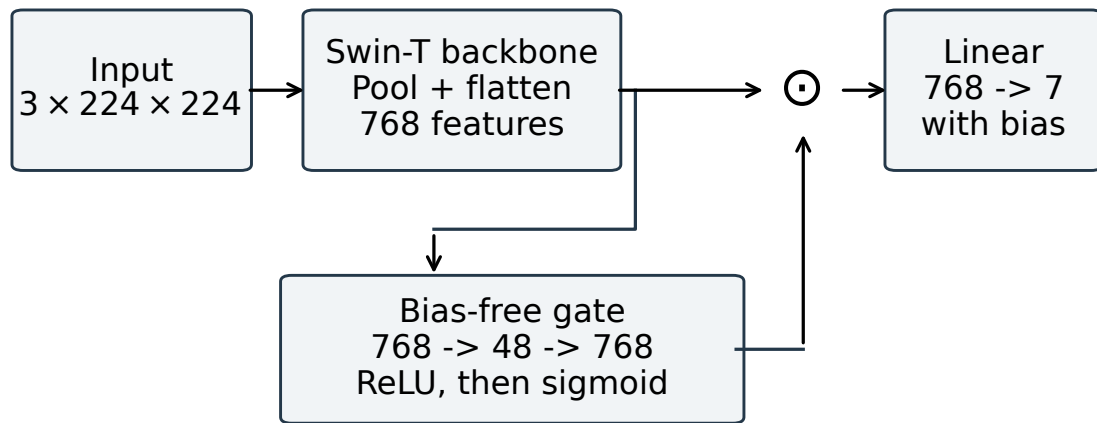
All numerical summaries are generated from `reported_results.csv`. S denotes Soft, U Uniform, H Hard, and T Tie-hard. Differences are percentage points. Three seeds are fixed, not samples supporting broad seed-population inference. The image source is reused; no external-domain or communication benefit is established. Global 80% refers to restricting a global accepted set to a stratum, whereas within-stratum 80% reranks that stratum independently. Supported-only risk conditions the vote reference on supported categories and is undefined at zero supported mass.

Table S1: Original four-condition paired seed contrasts at 80% coverage

State	Contrast	Seed 17	Seed 42	Seed 89
selected	Soft - Uniform	-0.3473	-0.4313	-0.3550
selected	Soft - Hard	-2.1908	-3.7328	-1.9084
selected	Tie-hard - Hard	-0.0153	-1.0992	-0.2977
selected	Soft - Tie-hard	-2.1756	-2.6336	-1.6107
epoch12	Soft - Uniform	-0.3282	-0.3931	-0.2290
epoch12	Soft - Hard	-1.1794	-1.4198	-1.4542
epoch12	Tie-hard - Hard	-0.2824	-0.5076	-0.4313
epoch12	Soft - Tie-hard	-0.8969	-0.9122	-1.0229

Table S2: Original four conditions: selected epochs from the common squared-distance criterion. Every fixed-epoch sensitivity checkpoint is epoch 12

Targets	Seed 17	Seed 42	Seed 89
Hard	3	2	5
Tie-hard	4	4	5
Uniform	9	6	10
Soft	9	10	10



One gate after pooling; 73,728 additional parameters

Figure S1: Fixed model head used in the five Swin conditions: one channel gate after pooled Swin-T features, followed by a seven-category linear classifier. This is not a new architecture claim

Local expression-cue prototype: illustrative controls

Synthetic text and score; not a participant record or measured prediction

Local camera preview No facial image shown Example: neutral Score: 0.62 (uncalibrated)	Compose a message I will send the draft tomorrow. ☐ Attach current expression cue Add to local preview
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Default: off; camera off or invalid/stale observation: omit the cue

Opt-in resets after each message. No face does not mean neutral.

Messages remain local; no recipient-side transport is implemented.

Figure S2: Synthetic illustration of the local sender-controlled interface. Text and category presentation are illustrative, not participant observations or a measured prediction. No recipient transport is implemented

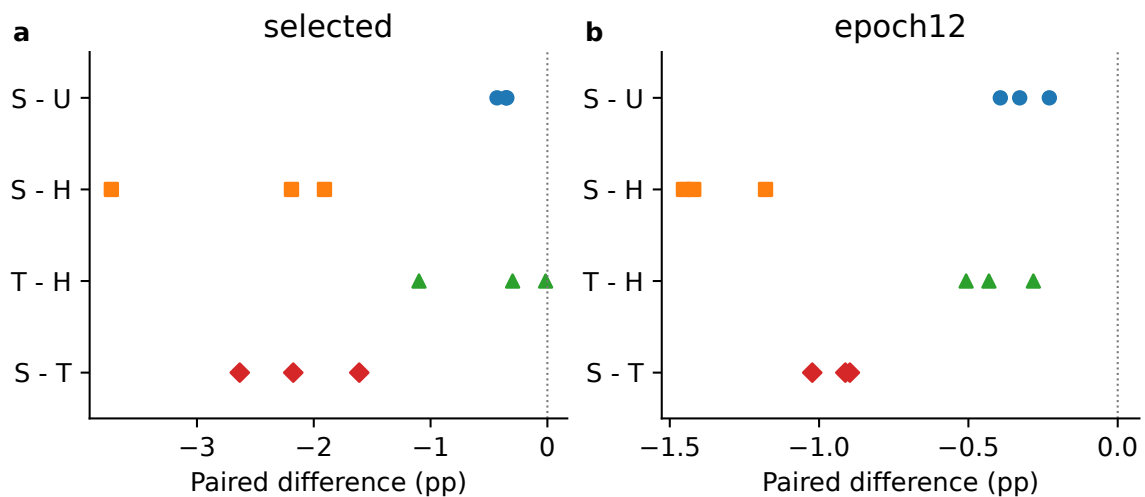


Figure S3: Original four conditions: individual paired seed differences at the common-coverage endpoint: (a) selected and (b) fixed epoch-12 weights. Exact values appear in Table S1; points can overlap

Table S3: Original four conditions, supported-mass strata: means over three selected-checkpoint seeds. Risk columns are percentages; accepted count can vary by seed

Targets	Stratum	Selection	N	Accepted	Risk	Supported-only
Hard	full	all	1732	1732.00	19.93	19.93
Hard	full	global80	1732	1497.33	14.96	14.96
Hard	full	within80	1732	1386.00	12.80	12.80
Hard	partial	all	1527	1527.00	47.53	38.44
Hard	partial	global80	1527	1116.33	41.08	31.30
Hard	partial	within80	1527	1222.00	42.69	33.04
Hard	zero	all	16	16.00	100.00	–
Hard	zero	global80	16	6.33	100.00	–
Hard	zero	within80	16	13.00	100.00	–
Tie-hard	full	all	1732	1732.00	19.42	19.42
Tie-hard	full	global80	1732	1507.00	14.51	14.51
Tie-hard	full	within80	1732	1386.00	12.33	12.33
Tie-hard	partial	all	1527	1527.00	46.84	37.52
Tie-hard	partial	global80	1527	1105.67	40.76	30.74
Tie-hard	partial	within80	1527	1222.00	42.23	32.38
Tie-hard	zero	all	16	16.00	100.00	–
Tie-hard	zero	global80	16	7.33	100.00	–
Tie-hard	zero	within80	16	13.00	100.00	–
Uniform	full	all	1732	1732.00	18.32	18.32
Uniform	full	global80	1732	1523.00	13.45	13.45
Uniform	full	within80	1732	1386.00	10.94	10.94
Uniform	partial	all	1527	1527.00	45.71	36.24
Uniform	partial	global80	1527	1090.00	38.40	28.24
Uniform	partial	within80	1527	1222.00	40.58	30.62
Uniform	zero	all	16	16.00	100.00	–
Uniform	zero	global80	16	7.00	100.00	–
Uniform	zero	within80	16	13.00	100.00	–
Soft	full	all	1732	1732.00	17.93	17.93
Soft	full	global80	1732	1529.00	13.23	13.23
Soft	full	within80	1732	1386.00	10.42	10.42
Soft	partial	all	1527	1527.00	45.36	35.75
Soft	partial	global80	1527	1080.00	37.71	27.35
Soft	partial	within80	1527	1222.00	40.00	29.77
Soft	zero	all	16	16.00	100.00	–
Soft	zero	global80	16	11.00	100.00	–
Soft	zero	within80	16	13.00	100.00	–

Full means all votes are supported; partial means positive supported and unsupported mass; zero means no supported votes. A risk of one in the zero stratum follows from the vocabulary, not a special model failure. Global selection may leave some strata empty; risk then remains undefined.

Table S4: Original four conditions, calibration half-split holdout-minus-fit risk gaps. Means and 5th/95th percentiles are descriptive percentage points over nonempty seed-split outcomes, not confidence intervals

Targets	Target	Nonempty	Mean gap	5th pct.	95th pct.
Hard	10%	300/300	-0.040	-2.215	+2.175
Hard	20%	300/300	+0.050	-2.328	+2.590
Hard	30%	300/300	+0.008	-2.752	+2.598
Tie-hard	10%	300/300	-0.002	-2.169	+2.399
Tie-hard	20%	300/300	+0.180	-2.344	+2.770
Tie-hard	30%	300/300	+0.072	-2.683	+2.859
Uniform	10%	300/300	+0.216	-1.808	+2.033
Uniform	20%	300/300	+0.193	-2.021	+2.617
Uniform	30%	300/300	+0.037	-2.442	+2.536
Soft	10%	300/300	+0.060	-1.871	+1.736
Soft	20%	300/300	+0.170	-2.004	+2.428
Soft	30%	300/300	+0.015	-2.410	+2.736

Each of 100 shared pixel-cluster splits uses disjoint calibration halves for fitting and checking the empirical threshold. Splits overlap one another. Minimum fitted accepted count is 100 and score ties are never split. The full ledger includes fit/holdout counts, risks, thresholds, and holdout coverage for every split, including empty policies.

Unconditional population composition

Calibration: unsupported mass 5.7247%, supported ambiguity 17.9720%, annotation floor 23.6967%.

Test: unsupported mass 7.8779%, supported ambiguity 18.7878%, annotation floor 26.6656%.

These annotation-only quantities do not depend on the fitted model. Accepted-set differences in the main figure condition on each model’s chosen threshold and can differ from these unconditional differences. None uniquely identifies a causal source of overshoot.

Reproduction map

REPRODUCIBILITY.md provides CPU numerical-audit and optional image-dependent inference commands. Online Resource 1 includes predictions, source hashes, category maps, original-row joins, all four adaptive protocols and tests. Online Resources 2–7 contain all selected and epoch-12 inference weights. No face images are redistributed. Internal automated verification is not external replication.

Prototype implementation boundary

Online Resource 1 provides a dimensioned model diagram (Fig. S1) and a synthetic interface illustration (Fig. S2); the latter is not a participant observation or a measured prediction. The application uses MediaPipe Face Detection with model selection 1 and minimum detection confidence 0.5 (MediaPipe Face Detection; main reference list). Exactly one detected face with a valid clipped crop is eligible. The crop then uses the same grayscale, resize, normalization, class order, and model head as the experiment. Benchmark images are already cropped and do not pass through the detector. Offline predictions use the recorded BF16 evaluation path; the application uses its documented FP32 inference path, so bitwise equality across paths is not assumed.

Predictions are requested no more often than once per second. A category can accompany a message only with sender opt-in, an active camera, and a valid observation at most 2.5 seconds old. Attachment defaults to off and resets after submission. No face, multiple faces, stale observations, and processing errors suppress the cue rather than becoming neutral. These are implementation choices, not empirically optimized safeguards. The interface labels its top-class score as uncalibrated. The confidence policies analyzed here remain offline; they are not presented as proven runtime protections.

The Streamlit/Python backend must run on the sender’s machine for local processing. The supplied launch configuration uses loopback, disables usage statistics, and omits external ICE servers. Frames are processed in memory without intentional application-level recording. Messages remain in session memory in a local preview; no recipient transport is implemented. The categorical cue can itself be sensitive. These boundaries do not establish a formal privacy guarantee or exclude operating-system and dependency behavior.

Code and numerical checks cover checkpoint metadata, gating placement, preprocessing, deterministic splits and augmentation, weighted losses, score ties, empty selections, metric definitions, and cue failure states. Runtime checks are software diagnostics, not a naturalistic usability study.

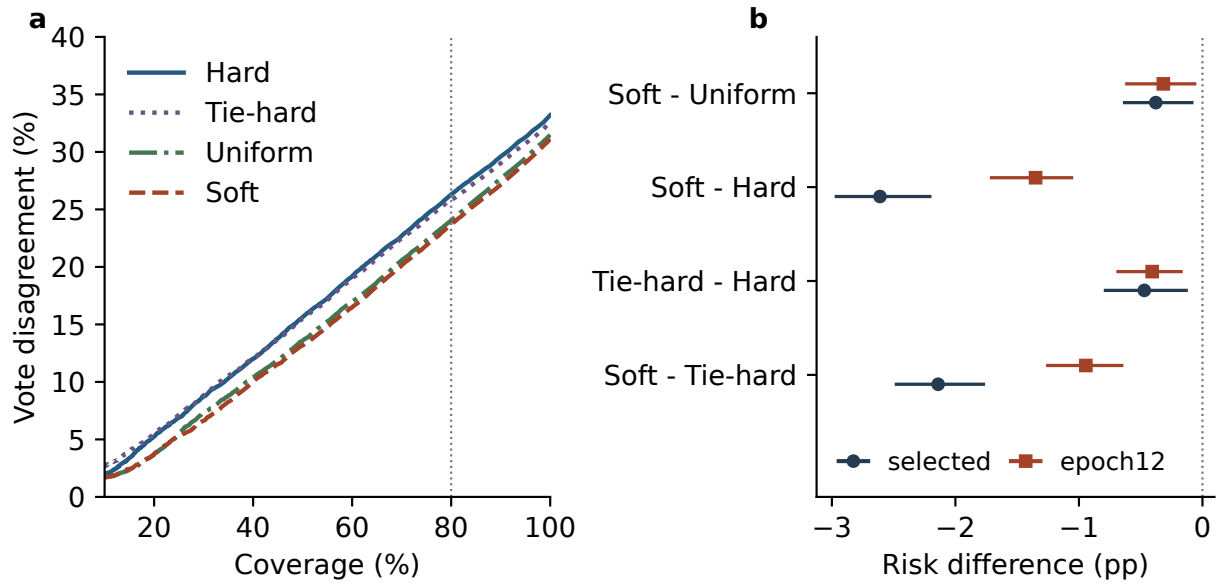
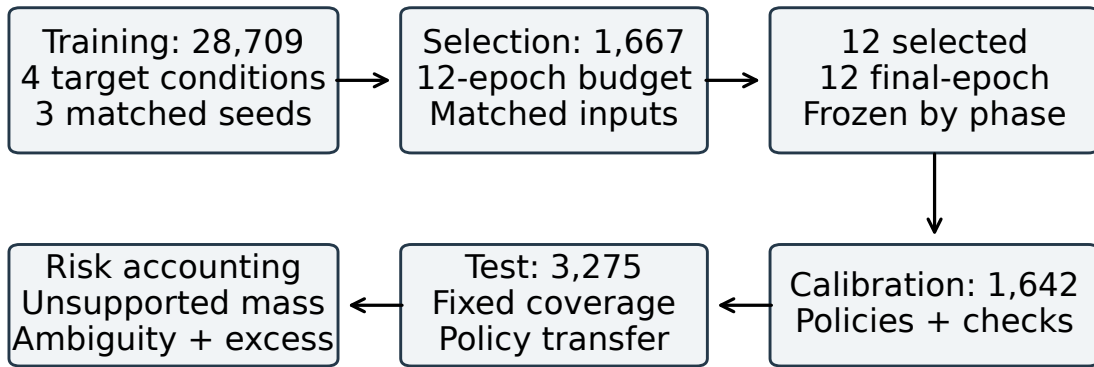


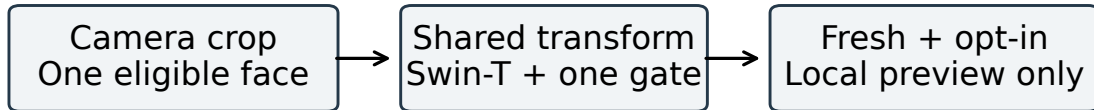
Figure S4: (a) Original four conditions: mean selected-checkpoint risk-coverage curves across three seeds; the displayed range begins at 10%, and every prefix is released. (b) Mean paired contrasts and conditional 95% pixel-cluster intervals, for selected and epoch-12 weights. Negative differences favor the first-named target. Intervals do not measure seed-population or external-domain uncertainty

a Matched offline experiment



Exact-pixel separation; not verified person-disjoint

b Secondary local prototype



No recipient transport or evaluated communication benefit

Figure S5: Historical four-condition offline flow and local application boundary; the main manuscript now isolates the experimental flow.

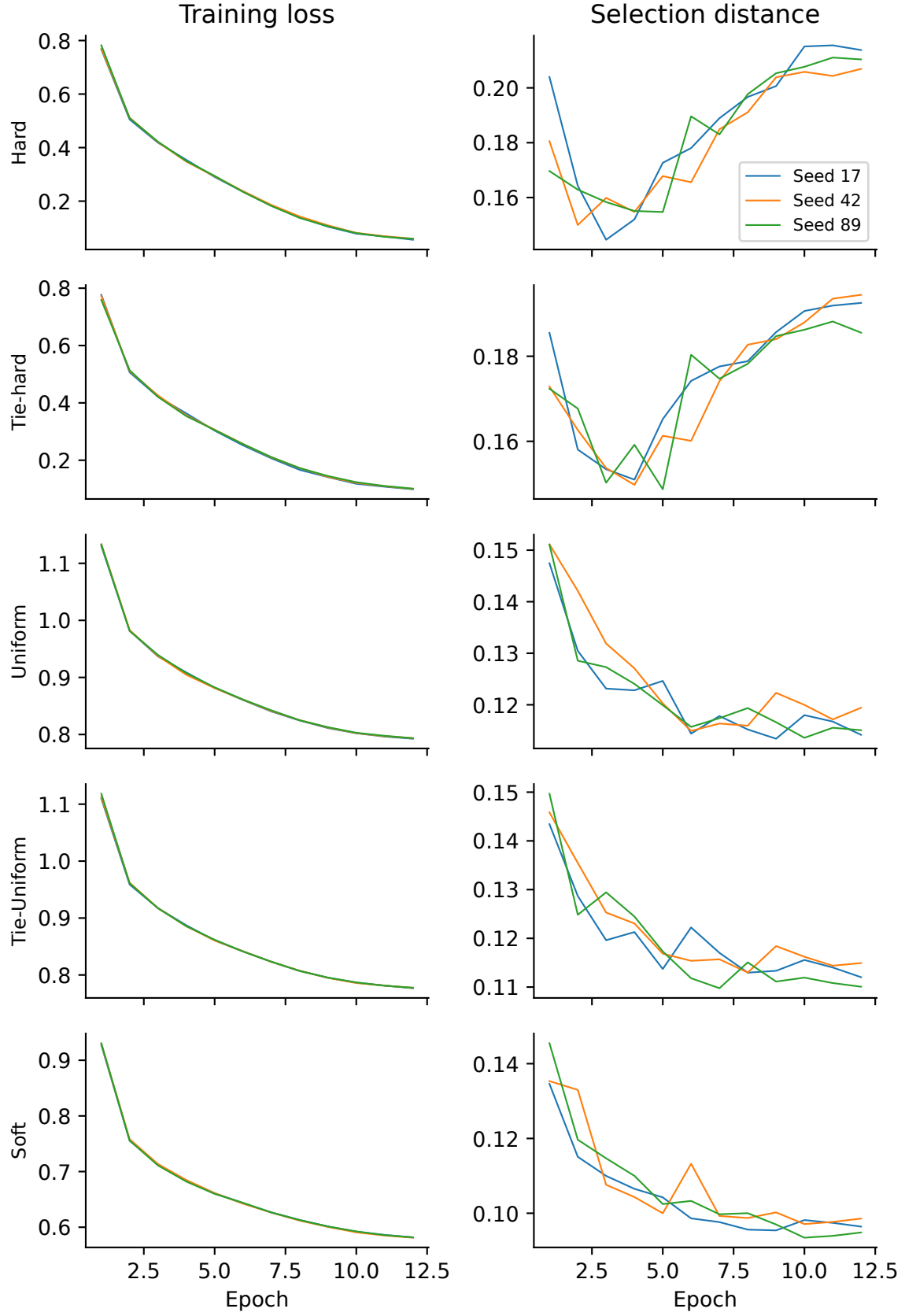


Figure S6: Training loss and selection-distance trajectories for the fifteen retained Swin runs. ResNet trajectories are supplied separately below. The target-dependent loss scales are not comparable performance endpoints.

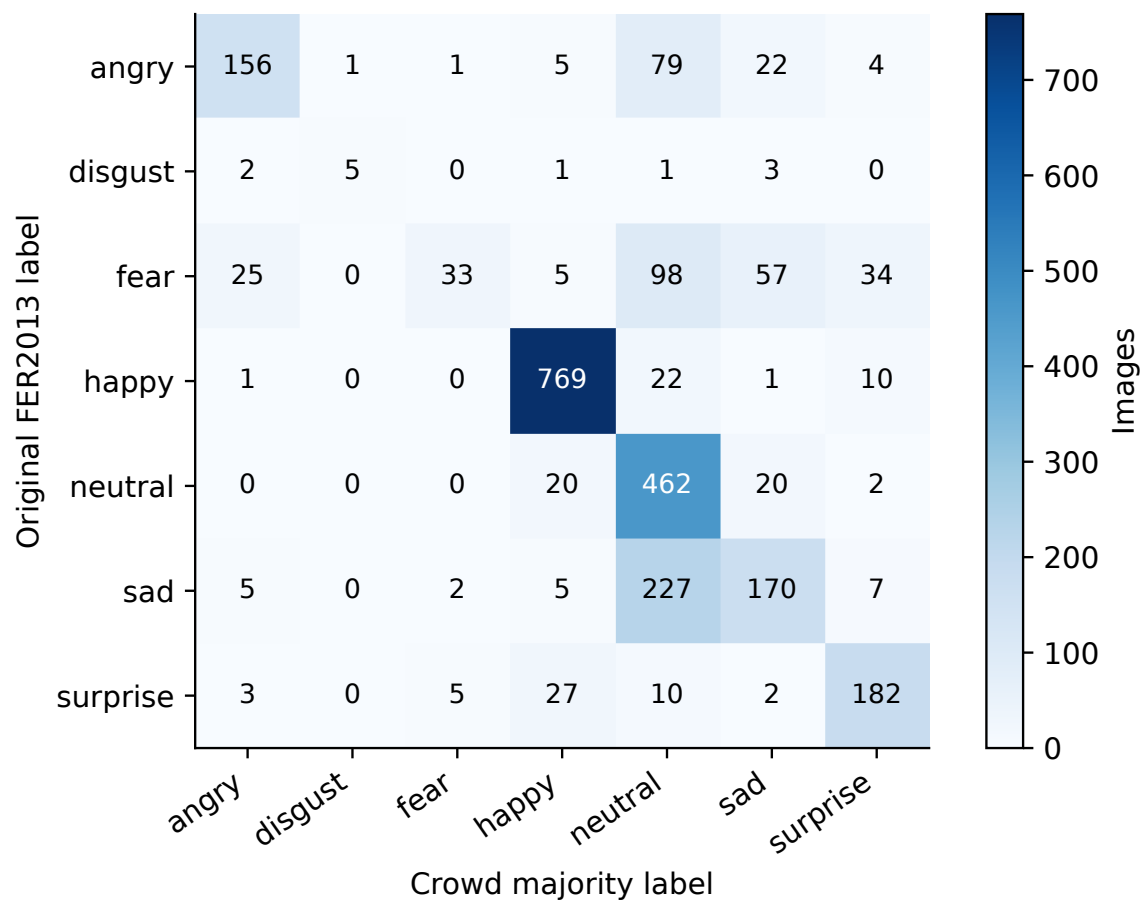


Figure S7: Original-label versus supported crowd-majority reference counts on exactly the same 2,484 primary-test images. This matrix compares references, not predictions.

Table S6: Secondary matched Tie-Uniform comparison at 80% coverage. Risks are percentages; differences and conditional 95% intervals are percentage points

State	Tie-Uniform	Soft	Difference	Interval
selected	24.272	23.684	-0.588	[-0.830, -0.281]
epoch12	23.939	23.632	-0.307	[-0.585, -0.024]

Table S7: Selected epochs under the common distribution-distance criterion

Targets	Seed 17	Seed 42	Seed 89
Hard	3	2	5
Tie-hard	4	4	5
Uniform	9	6	10
Tie-Uniform	12	8	7
Soft	9	10	10

Unsupported category vote mass

Targets	Population	Scope	Contempt	Unknown	Not-a-face
Hard	calibration	all	1.797	3.666	0.262
Hard	calibration	global80	1.525	3.092	0.185
Hard	test	all	1.554	5.667	0.656
Hard	test	global80	1.291	4.966	0.391
Tie-hard	calibration	all	1.797	3.666	0.262
Tie-hard	calibration	global80	1.446	3.064	0.274
Tie-hard	test	all	1.554	5.667	0.656
Tie-hard	test	global80	1.302	4.931	0.433
Uniform	calibration	all	1.797	3.666	0.262
Uniform	calibration	global80	1.329	2.877	0.193
Uniform	test	all	1.554	5.667	0.656
Uniform	test	global80	1.224	4.765	0.421
Tie-Uniform	calibration	all	1.797	3.666	0.262
Tie-Uniform	calibration	global80	1.314	2.943	0.238
Tie-Uniform	test	all	1.554	5.667	0.656
Tie-Uniform	test	global80	1.246	4.821	0.405
Soft	calibration	all	1.797	3.666	0.262
Soft	calibration	global80	1.291	3.004	0.241
Soft	test	all	1.554	5.667	0.656
Soft	test	global80	1.228	4.749	0.571

Entries are mean vote-mass percentages across three seeds, not proportions of images with that label. All-image masses are model-independent; global80 conditions on each model’s ranking. The three categories do not share one semantic interpretation.

Per-seed fixed-image cross-evaluation

State	Predictor	Seed	Uniform set	Soft set
selected	Uniform	17	24.0496	24.0878
selected	Soft	17	24.1145	23.7023
selected	Uniform	42	24.1718	24.1908
selected	Soft	42	24.2023	23.7405
selected	Uniform	89	23.9656	23.8321
selected	Soft	89	24.1260	23.6107
epoch12	Uniform	17	23.9733	24.0076
epoch12	Soft	17	23.9389	23.6450
epoch12	Uniform	42	24.0840	23.8817
epoch12	Soft	42	24.2595	23.6908
epoch12	Uniform	89	23.7901	23.6489
epoch12	Soft	89	23.9122	23.5611

Risk entries are complete-vote disagreement percentages at exactly 2,620 accepted images.

State	Seed	Intersection	Union	Jaccard	Overlap fraction
selected	17	2397	2843	0.8431	0.9149
selected	42	2381	2859	0.8328	0.9088
selected	89	2402	2838	0.8464	0.9168
epoch12	17	2394	2846	0.8412	0.9137
epoch12	42	2391	2849	0.8392	0.9126
epoch12	89	2404	2836	0.8477	0.9176

Overlap fraction divides the intersection by 2,620, whereas Jaccard divides by the union. The two accepted sets can share most images while their differing tails change mean risk.

Second-backbone robustness diagnostics

Table S10: Within-backbone Soft-minus-control disagreement at 80% coverage (percentage points). Intervals condition on fitted models, three seeds, observed pixels and votes

Backbone	State	Control	Difference	95% interval
Swin-T	selected	Uniform	-0.378	[-0.644, -0.073]
Swin-T	selected	Tie-Uniform	-0.588	[-0.830, -0.281]
Swin-T	epoch12	Uniform	-0.317	[-0.626, -0.050]
Swin-T	epoch12	Tie-Uniform	-0.307	[-0.585, -0.024]
ResNet-18	selected	Uniform	+0.173	[-0.167, +0.455]
ResNet-18	selected	Tie-Uniform	+0.042	[-0.258, +0.343]
ResNet-18	epoch12	Uniform	+0.170	[-0.157, +0.467]
ResNet-18	epoch12	Tie-Uniform	+0.190	[-0.126, +0.515]

ResNet comparisons are adaptive secondary robustness checks, not independent-image confirmation or architecture comparisons.

Table S11: ResNet-18 complete-vote risk (%) at 80% coverage, means across paired seeds

Targets	Selected	Epoch 12
Soft	24.700	24.603
Uniform	24.527	24.433
Tie-Uniform	24.658	24.413

Table S12: ResNet-18 global 80% accepted counts in full/partial/zero-supported-mass strata. Entries list seeds 17, 42, 89

Targets	Full	Partial	Zero
Soft	1509, 1533, 1522	1106, 1077, 1088	5, 10, 10
Uniform	1520, 1528, 1522	1091, 1085, 1092	9, 7, 6
Tie-Uniform	1523, 1521, 1511	1088, 1090, 1101	9, 9, 8

Each seed retains 2,620 of 3,275 test images. Zero-mass population size is 16.

selected fixed-image comparisons

Control	Predictor	Selector	Seed 17	Seed 42	Seed 89
Uniform	Soft	Soft	24.8321	24.7824	24.4847
Uniform	Soft	Uniform	25.1183	25.0382	24.7595
Uniform	Uniform	Soft	25.2214	25.0687	25.0458
Uniform	Uniform	Uniform	24.4847	24.6450	24.4504
Tie-Uniform	Soft	Soft	24.8321	24.7824	24.4847
Tie-Uniform	Soft	Tie-Uniform	25.3588	25.3397	25.1145
Tie-Uniform	Tie-Uniform	Soft	24.9656	24.7748	24.4313
Tie-Uniform	Tie-Uniform	Tie-Uniform	24.7137	24.6679	24.5916

Entries are complete-vote disagreement percentages at 2,620 retained images.

epoch12 fixed-image comparisons

Control	Predictor	Selector	Seed 17	Seed 42	Seed 89
Uniform	Soft	Soft	24.6069	24.6450	24.5573
Uniform	Soft	Uniform	24.8702	24.7519	24.9313
Uniform	Uniform	Soft	24.9275	24.8053	24.9847
Uniform	Uniform	Uniform	24.5344	24.1565	24.6069
Tie-Uniform	Soft	Soft	24.6069	24.6450	24.5573
Tie-Uniform	Soft	Tie-Uniform	24.7176	24.9618	24.9504
Tie-Uniform	Tie-Uniform	Soft	24.6412	24.8779	24.6756
Tie-Uniform	Tie-Uniform	Tie-Uniform	24.2824	24.5267	24.4313

Entries are complete-vote disagreement percentages at 2,620 retained images.

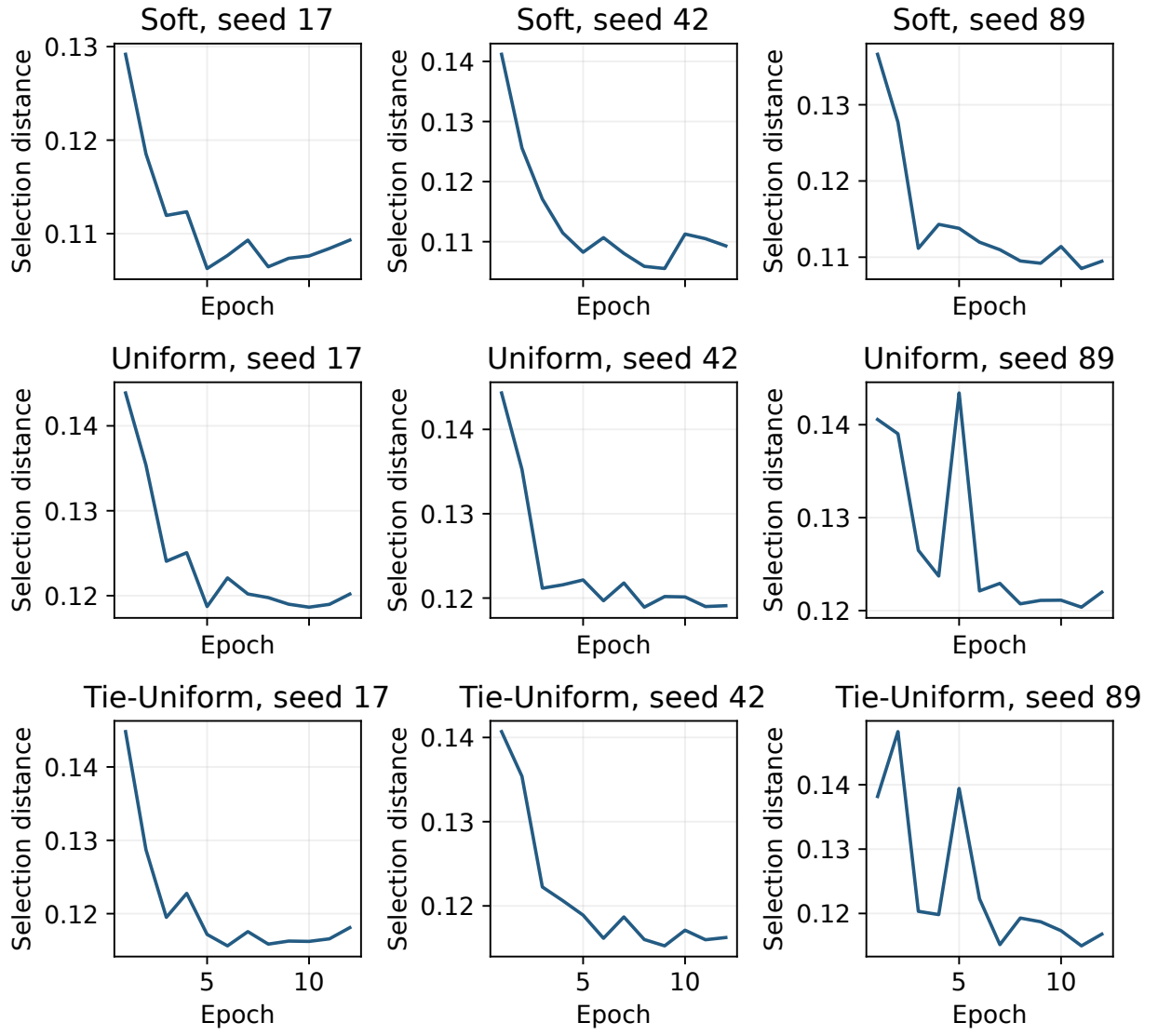


Figure S8: Selection-distance trajectories for the nine ResNet runs. Selected and epoch-12 models are both evaluated; these curves do not establish convergence

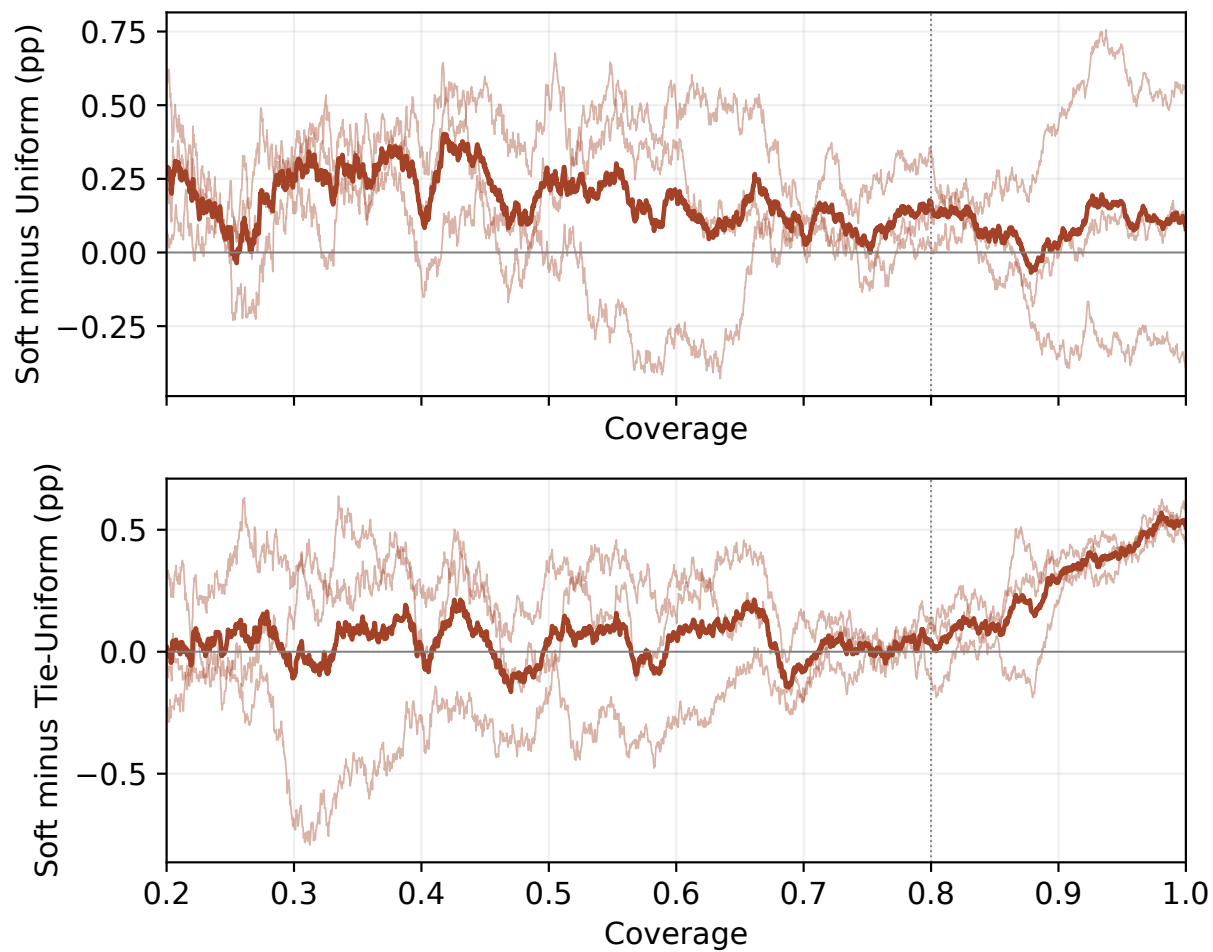


Figure S9: ResNet selected-model own-ranking risk differences at 20–100% common coverage. Thin curves are paired seeds and thick curves their mean. The dotted marker identifies the prespecified 80% secondary endpoint. These descriptive curves have no simultaneous uncertainty bands and do not compare predictions on identical accepted images

ResNet reference-specific weighted F1

State	Targets	Original/all	Original/subset	Crowd/subset
selected	Soft	0.5564	0.6508	0.9053
selected	Uniform	0.5558	0.6531	0.9093
selected	Tie-Uniform	0.5588	0.6572	0.9181
epoch12	Soft	0.5612	0.6561	0.9110
epoch12	Uniform	0.5563	0.6542	0.9135
epoch12	Tie-Uniform	0.5608	0.6597	0.9209

All images: 3,275; identical majority subset in the last two columns: 2,484. No extra relabeling is performed. Per-seed unsupported masses and overlaps are supplied in the CSV ledger.